Primary Research Question (Descriptive):

To what extent can LLM-based data agents accurately diagnose the underlying missing data mechanisms (MCAR, MAR, MNAR) and autonomously apply statistically appropriate imputation methods when presented with incomplete datasets?

Seed References & Prior Work

Donoho (2017): Highlights the central role of statistical thinking in data science, emphasizing that data analysis is not merely about computation but fundamentally about understanding underlying assumptions. This principle grounds the project's focus on measuring an agent's "assumption-awareness" rather than just output correctness.

Sun et al. (2025, Survey on LLM-based Agents): Provides a comprehensive overview of LLM agents in data science tasks. Crucially, this survey identifies a major gap that this project aims to fill: current evaluations predominantly focus on mechanical task performance, leaving deeper statistical reasoning largely unassessed.

LLMs for Missing Data Imputation (2026): Investigates the specific behavioral quirks of LLMs in missing data tasks, including hallucination effects and logical inconsistencies. This literature empirically supports the project's core hypothesis: generating a "correct-looking" imputed number does not necessarily imply that the agent possesses correct statistical understanding.

Jäger et al. (2021, A Benchmark for Data Imputation Methods): Establishes a comprehensive benchmark framework for evaluating data imputation algorithms by systematically inducing various missingness mechanisms (MCAR, MAR, MNAR) at different ratios across multiple datasets. This paper provides the direct methodological blueprint and rigorous justification for my automated data generation pipeline (using artificial amputation), ensuring the evaluation testbed is standardized and possesses absolute ground truth.

Planned Paper Outline

1. Abstract (~0.25 pages)Concisely summarize the problem, benchmark approach, key findings, and implications.
2. Introduction (~1 page)Establish the context, gap, and contribution. Explicitly state research questions and outline the paper's structure.
3. Background and Related Literature (1-2 pages)Thematically synthesize substantive and methodological literature, avoiding one-by-one summaries.
4. Research Design and Methods (1-2 pages)Detail dataset provenance and analytical methods (RMSE, LLM-as-a-Judge). Justify methodology and explicitly disclose any AI tool usage.
5. Results (2-4 pages)Objectively report trends, effect sizes, and null results using tables/figures without premature interpretation.
6. Discussion (1.5-3 pages)Interpret findings against theoretical framing and discuss implications for human-AI trust.
7. Limitations and Future Research (0.5-1 page)Critically evaluate internal, external, and evaluation validity. Propose concrete future research directions.

8. Conclusion (0.5-1 page)Synthesize contributions without introducing new claims.